

HPTQ: Hardware-Friendly Post Training Quantization

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Abstract

Neural network quantization enables the deployment of models on edge devices. An essential requirement for their hardware efficiency is that the quantizers are hardware-friendly: uniform, symmetric and with power-of-two thresholds. To the best of our knowledge, current post-training quantization methods do not support all of these constraints simultaneously. In this work we introduce a hardware-friendly post training quantization (HPTQ) framework, which addresses this problem by synergistically combining several known quantization methods. We perform a large-scale study on four tasks: classification, object detection, semantic segmentation and pose estimation over a wide variety of network architectures. Our extensive experiments show that competitive results can be obtained under hardware-friendly constraints.

1 Introduction

Deep neural networks have shown state-of-art performance in many real-world computer vision tasks, such as image classification [1, 2], object detection [3, 4, 5], semantic segmentation [6] and pose estimation [7, 8]. However, the deployment of deep neural networks on edge devices is still considered a challenging task due to limitations on available memory, computational power and power consumption.

Quantization [9] is a common approach to tackle this challenge with minimal performance loss, by reducing the bit-width of network weights and activations. Quantization methods can be roughly divided into two categories: quantization aware training (QAT) and post-training quantization (PTQ). QAT methods [10, 11, 12, 13] retrain the network in order to recover the accuracy degradation

An implementation of this work is available at https://github.com/sony/model_optimization.